Measuring the severity of stress scenarios in nats: what entropy offers to stress-test designers

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Abstract

Authorities that design stress scenarios, such as the adverse scenario of the EU-wide stress test coordinated by the European Banking Authority (EBA), must answer two questions: is the scenario severe enough, and is it still plausible? This paper proposes to measure both on a single scale, entropy, expressed in nats. A nat is a quantity of information. For a pure shift of the mean, a scenario of η nats is a shock of $\sqrt{2\eta}$ standard deviations: one nat buys 1.4 standard deviations. We apply this scale to 33 years of returns on twenty US stocks, to Kenneth French's 48 industry portfolios, and to a euro-area panel covering the five segments of the ECB's Composite Indicator of Systemic Stress (CISS). Three results stand out. **(1)** A simple robustness test, which reruns every computation on data whose time order has been shuffled, removes most of the crisis signals that entropy appears to show: they are artefacts of estimation. One fact survives on every panel: under stress, markets couple, meaning that the number of independent sources of risk falls. **(2)** Measured in nats, classical scenarios that shock volatilities, correlations and levels separately pay 15 to 33 % too much for their tail loss, and the split of the shock between volatility and correlation can be calibrated on history instead of being assumed. **(3)** The coupling measure decomposes exactly across market segments. In the euro area it flags recessions as well as the CISS does and shows which segments stress ties together, but it does not forecast growth.

1 Introduction

Two facts about financial prices sit uneasily together. The first is that the Gaussian model is the natural answer to a question of ignorance: fix a mean and a variance, know nothing else, and the maximum-entropy distribution is Gaussian. The second is that returns are not Gaussian: they have fat tails, negative skewness, and their dependence tightens precisely when it matters.

The usual reaction treats the second fact as a defect of the first model and patches it: add jumps to the price, add a stochastic volatility factor, add a copula. This paper goes the other way. If the Gaussian is what maximum entropy delivers, then any departure from the Gaussian measures how far the market is from maximal ignorance, that is, a measure of information. The question is therefore not how to patch the Gaussian, but how to read the size, dynamics and direction of the departure.

The paper is organised around an asymmetry in that departure. Information arrives in bursts and is absorbed gradually. A central-bank surprise, a default, a war removes uncertainty in an instant; the market then spends weeks working out the implications, and uncertainty rebuilds. If entropy is the right currency of uncertainty, its dynamics should be a sawtooth: sharp drops, slow recoveries. Half of that shape is a theorem: receiving information cannot, on average, increase uncertainty. The other half, the slow recovery, is a hypothesis that we test.

Who this paper is for. It is written first for institutions that design stress scenarios for others: the European Systemic Risk Board (ESRB) and the ECB, which write the adverse scenario of the EBA stress test, national central banks, the IMF in its financial sector assessment programmes, and insurance supervisors. These institutions must produce scenarios that are "severe but plausible" (BCBS, 2018). In practice severity is measured in losses and plausibility is judged by experts. A common unit for both is missing. Entropy provides one: it measures in nats the distance between a scenario and the current state of the market, whatever the asset class, and it reduces to standard deviations in simple cases. Judging the plausibility of a scenario by a statistical distance is not new (Breuer et al., 2009; Breuer & Csiszár, 2013; Glasserman & Xu, 2014). What we add is a calibration of that distance on data, a test that separates real signals from artefacts, and an application to the euro area.

Four contributions.

1. *A measurement framework* (section 3). The uncertainty of a return vector splits exactly into three channels: scale (volatilities), dependence (correlations) and shape (skewness and fat tails). Removing scale leaves a unit-free index, the structural index $\mathcal{J}$, which measures the market's distance from maximal ignorance.
2. *A robustness test* (sections 4 and 5). An entropy measure estimated on real data jumps up and decays slowly even when no information arrives: this is a mechanical effect of estimation. We therefore judge every result against *placebo nulls*, that is, against the same computations run on resampled data. Only one result survives everywhere: under stress, markets couple.
3. *A severity unit for scenarios* (section 6). A scenario of severity η is the worst distribution within η nats of the current distribution. The radius η is calibrated on how fast the market itself revises its distribution. The same unit can audit an existing scenario and choose how to split the shock between volatility and correlation.
4. *A monitoring indicator* (section 7). The coupling measure decomposes exactly across market segments. Applied to the five segments of the ECB's CISS, it shows which segments stress ties together, which directly informs the correlation structure of a scenario.

A warning. The structural index is computed from estimated moments (variances, correlations, skewness, kurtosis). On fat-tailed data, estimation noise pushes such an index *up* abruptly and lets it come down *slowly*. That is exactly the sawtooth the theory predicts. The result is uncomfortable and, we think, the most useful thing in the paper: most of the signals that look overwhelming do not survive the null. Anyone building a crisis indicator from estimated moments should run this null before believing their own chart.

Outline. Section 2 defines the concepts used. Section 3 presents the framework and the eight hypotheses tested. Section 4 describes the data and the placebo test, section 5 the results. Sections 6 and 7 present the two applications: measuring scenario severity and the entropic CISS. Section 8 summarises recommendations and limitations. Technical justifications are in the appendix; full proofs are in a separate theoretical supplement.

2 The concepts used

This section defines, in the order they are needed, every concept the rest of the paper uses. No calculation is needed to follow what comes next.

Entropy. The entropy $h(p)$ of a distribution p measures its degree of uncertainty: the more spread out the distribution, the higher the entropy. For a continuous variable, $h(p) = -\int p \log p$. Its level depends on units (expressing returns in percent rather than decimals shifts it by a constant): only its *changes* and the *divergences* below have absolute meaning.

Nat. The unit of information obtained with the natural logarithm. 1 nat ≈ 1.44 bit. All quantities in the paper are in nats.

Maximum entropy. Among all distributions with a given mean and covariance, the Gaussian has maximal entropy: it is the distribution that assumes the least beyond those two moments. Any departure from the Gaussian is therefore additional information.

Kullback–Leibler (KL) divergence. $D_{KL}(p \,\|\, q) = \int p \log(p/q) \geq 0$ measures in nats the gap between a distribution p and a reference distribution q. It is zero if and only if $p = q$ and does not depend on units. It is not a distance (it is not symmetric), hence the word "divergence". Practical reading: if two Gaussians share the covariance Σ and have means μ_0 and μ_1, $D_{KL}(p_1 \,\|\, p_0) = \frac{1}{2}(\mu_1 - \mu_0)^\top \Sigma^{-1}(\mu_1 - \mu_0)$, half the squared Mahalanobis distance. A shock of k standard deviations therefore costs $k^2/2$ nats.

Mutual information. $I(X;Y)$ is the KL divergence between the joint law of X and Y and the product of their separate laws. It measures what X and Y share and is zero if they are independent. For two Gaussian variables with correlation ρ, $I = -\frac{1}{2}\log(1-\rho^2)$: 0.83 nat for $\rho = 0.9$.

Total correlation (coupling). For N assets with correlation matrix R, $C(R) = -\frac{1}{2}\log\det R \geq 0$ extends mutual information to more than two variables (Watanabe, 1960). It is zero if the assets are independent and rises when they start to move together. We call it *coupling*. An equivalent reading is the *effective number of risk modes*, which equals N when assets are independent and tends to 1 when a single common factor explains everything.

The three entropy channels. The entropy of a return vector splits exactly into three terms (section 3): *scale* (the level of volatilities), *dependence* (coupling, counted negatively) and *shape* (the deficit defined below).

Entropy deficit $\mathcal{D}$. The KL divergence between the true distribution and the Gaussian with the same mean and covariance. It measures what the Gaussian misses: skewness and fat tails.

Structural index $\mathcal{J}$. The sum of shape and dependence, $\mathcal{J} = \mathcal{D} + C$. It is the KL divergence between the true distribution and independent Gaussians with the same volatilities. It is zero if returns are Gaussian and independent, and it does not depend on the level of volatility.

Stress regime. Dates on which 21-day market volatility exceeds the threshold of the top 10 % observed *up to that date*. The threshold never uses future information.

Placebo null. The same computations, rerun on data whose time order has been shuffled. The *i.i.d.* null draws days one by one: it keeps fat tails and cross-asset correlation, and destroys all dynamics. The *block* null draws 21-day blocks: it also keeps the clustering of turbulent days, and destroys only longer-run dynamics. Each null is repeated 200 times. A result *passes* a null if at most 5 % of the 200 repetitions produce a result as strong as the real data.

Information flow. $I_t^{(h)} = D_{KL}(p_t \,\|\, p_{t-h})$: by how many nats the market has revised its own distribution over h days.

Scenario of severity η and return period. The worst distribution within η nats of the current distribution. It is assigned the return period of an information flow of the same size: a radius the market travels in one month only once a decade is a "ten-year" scenario.

Tail loss ($\mathbf{ES}_{99}$). The 99 % expected shortfall: the average loss on the worst 1 % of days.

3 The framework: from maximal ignorance to three channels

3.1 Why the Gaussian is the reference

The step from "maximum entropy gives a Gaussian" to "prices are Gaussian" is often asserted and rarely argued, because maximum entropy takes the variance as given without saying where it comes from. Appendix A closes that bridge. If time is measured in traded volume rather than in days, volume fixes the variance, and the least committal distribution of the price increment is Gaussian. This is the mixture-of-distributions hypothesis (Clark, 1973; Ané & Geman, 2000), derived rather than assumed. In calendar time, returns depart from the Gaussian for two reasons: the volume clock is random, or information constrains the market. No empirical result in the paper depends on this bridge; it motivates the Gaussian as the zero point.

3.2 Three channels and the structural index

Write the covariance $\Sigma = DRD$, where D holds the volatilities σ_i and R the correlations. The entropy of the true distribution p splits exactly (appendix E):

$$h(p) \;=\; \tfrac{N}{2}\log(2\pi e) \;+\; \underbrace{\textstyle\sum_i \log \sigma_i}_{\text{scale}} \;+\; \underbrace{\tfrac{1}{2}\log\det R}_{\text{dependence } (=-C)} \;-\; \underbrace{\mathcal{D}}_{\text{shape}}. \tag{1}$$

The first two terms form the entropy of the Gaussian with the same covariance, the ceiling. The deficit $\mathcal{D}$ says how far below it the true distribution lies.

The scale channel is a problem: it rises with volatility. In a crisis, entropy computed from the covariance matrix alone *rises* (by 5 nats on our main panel), whereas intuition says a crisis is a moment when information constrains the market. Using the level of Gaussian entropy as a stress indicator therefore gets the sign wrong. We remove that channel and keep the other two:

$$\mathcal{J} \;=\; \mathcal{D} \;+\; C \;=\; D_{KL}\!\left(p \,\|\, \textstyle\prod_i q_i\right) \;\geq\; 0, \tag{2}$$

where the q_i are Gaussians with each asset's volatility. $\mathcal{J}$ is a genuine divergence (appendix E): it is positive, zero if and only if returns are Gaussian and independent, and unit-free. It rises when assets couple (C) or when their distributions depart from the Gaussian ($\mathcal{D}$), and one can always say which part of a move comes from each channel.

3.3 What the framework predicts, and the eight hypotheses

News reduces uncertainty. Formally, receiving a signal Y lowers the *average* entropy of the return X by exactly the mutual information $I(X;Y)$ (Cover & Thomas, 2006). The entropy jump therefore has a size measurable in nats: the information content of the news. The framework adds two postulates, detailed in appendix B. The first (*relaxation*) says that, absent new information, $\mathcal{J}$ slowly returns to its normal level. The second (*sign*) says that jumps come with negative skewness, because $\mathcal{J}$ alone cannot tell a crash from a rally. The result is a sawtooth: $\mathcal{J}$ rises by jumps and

comes down slowly. The framework also predicts that important news, because it hits every asset on the same day, raises coupling and tail thickness together.

Table 1 turns these predictions into eight testable hypotheses.

	Statement
H1	*Regime signature.* Under stress, coupling rises, the shape deficit rises, hence $\mathcal{J}$ rises.
H2	*Compensation.* When volatilities rise, dependence absorbs part of the entropy increase (slope $\beta > 0$).
H3	*One-sided jumps.* Large moves of $\mathcal{J}$ are increases, small ones decreases: upward episodes are more numerous and larger than downward ones.
H4	*Relaxation.* After an episode, $\mathcal{J}$ decays with a half-life longer than the estimator's own memory.
H5	*Sign.* Jumps and upward episodes of $\mathcal{J}$ coincide with negative market skewness.
H6	*Channel coupling.* Dependence and tails move independently in calm markets and together under stress.
H7	*Forecasting.* Today's entropy improves, out of sample, the forecast of risk over the next 21 days beyond volatility.
H8	*Crisis composition.* In crisis episodes, dependence is the dominant channel more often than in data without dynamics.

Table 1: The eight hypotheses tested. H3, H4, H5 and H8 concern jumps: they are tested day by day and by *episode*, an episode merging jumps less than 40 days apart.

4 Data and method

4.1 Data

- *Main panel:* twenty individual S&P 500 stocks, daily, 1990–2022 (8,312 days), covering the Asian crisis, LTCM, the dot-com bubble, 2008, the sovereign debt crisis, COVID-19 and 2022. Individual stocks rather than portfolios, because sector aggregation mechanically creates dependence.
- *Industry panel:* Kenneth French's 48 daily industry portfolios, 1990–2026 (about 9,000 days).
- *Control panel:* the DAX, SMI, CAC and FTSE indices, 1991–1998 (Bollerslev & Ghysels, 1996), too small to carry a result on its own.
- *Euro-area panel:* 28 daily series across the five CISS segments, 2005–2026 (section 7).

4.2 Estimation

Correlations and volatilities are estimated on a 252-day rolling window with the Ledoit & Wolf (2004) correction, or with exponential weighting (60-day half-life) for everything to do with jumps. The shape deficit is estimated asset by asset with the Hyvärinen (1998) approximation, accurate to a few percent on fat-tailed laws like those of daily returns. A jump is flagged when the daily change in $\mathcal{J}$ exceeds four times a local scale that is robust to jumps (Barndorff-Nielsen & Shephard, 2004). Regime comparisons use a 252-day block bootstrap; forecasts are evaluated out of sample with the Diebold & Mariano (1995) test. Appendix D gives the details.

4.3 Why a placebo null is indispensable

The index $\mathcal{J}$ is a convex, positive function of estimated moments. This creates a trap. When an extreme day enters the estimation window, the index jumps up; it then comes down slowly as that day loses weight. This movement mimics exactly the sawtooth the theory predicts, without any information having arrived.

A simulation shows it. Feed independent Gaussian returns with constant covariance through the same pipeline: a world without any dynamics. A single injected stress day raises $\mathcal{J}$ by 2.1 nats, and the index decays with a half-life of 35 days, through estimator memory alone. With no event at all, the detector finds 37 positive jumps and no negative one.

A result is therefore evidence only if it exceeds what data without dynamics produce. Hence the two nulls defined in section 2. They run exactly the same pipeline, including the definition of the stress regime. They answer two different questions. The i.i.d. null asks: beyond the fact that turbulent periods contain large days, does the measure detect anything? The block null asks the harder question: beyond the mere clustering of turbulent days, does it detect dynamics specific to crises? A result that passes both is established. A result that passes only one is not identified.

5 Results: what survives the null

Table 2 summarises the verdicts. Detailed figures are in appendix D (tables 10 and 11).

	Raw result (S&P 500)	Verdict S&P 500	Verdict FF49 industries
H1 coupling	rise under stress +0.99 nat	**passes both**	**passes both**
H1 index $\mathcal{J}$	+1.93 nat	fails both	fails both
H1 shape $\mathcal{D}$	+0.27 nat	null stronger	null stronger
H2 compensation	$\beta = 0.20$ under stress	fails both	i.i.d. only
H3 jumps, daily	158 up, 9 down	null stronger	fails both
H3 jumps, by episode	65 up, 7 down	null stronger	not tested
H3 largest episode	15.8 nats (COVID)	**passes both**	not tested
H4 half-life, daily	121 days	not identified	not identified
H4 half-life from peak	18 days (74 for the 5 largest)	fails both	not tested
H4 range of $\mathcal{J}$	18.95 nats	**passes both**	i.i.d. only
H5 sign, daily	correlation −0.24	fails ($p = 0.08$–0.10)	fails both
H5 sign, by episode	skewness falls in 62 %	i.i.d. only	not tested
H6 channel coupling	0.04 calm, 0.63 stress	null stronger	null stronger
H7 forecasting	gain $\leq 3\,\%$, $p = 0.23$	rejected	—
H8 composition	dependence dominant 46 %	i.i.d. only	i.i.d. only

Table 2: Verdicts against the two nulls (200 repetitions each). "Null stronger": resampled data produce a larger effect than the real data. For H7, the test is out of sample and covers the main and control panels.

5.1 The fact that survives: under stress, markets couple

Under stress, coupling among the twenty stocks rises by 0.99 nat. The i.i.d. null produces only 0.21 nat (none of the 200 repetitions reaches the observed value), the block null 0.51 nat ($p = 0.03$). Concretely, the effective number of independent risk sources falls from 13.1 to 11.4 out of 20. The same result holds on French's 48 industries: 3.65 nats against 0.56 and 2.46 ($p = 0.025$). It is the

only result common to both panels. Its relative strength is similar on individual stocks and on industries, which rules out an aggregation artefact.

The second result that passes both nulls on the main panel is the *range* of $\mathcal{J}$: the index travels 18.95 nats over the sample, against 10.05 and 13.77 in the nulls ($p = 0.035$ against blocks). Real markets move the index further than any resampling of the same returns. This result does not, however, replicate against the block null on industries.

A jump is not an event. The detector judges *daily* changes in $\mathcal{J}$. A crisis lasting several weeks therefore produces several jumps: the COVID crisis yields nine between 24 February and 16 March 2020, while the index rises by 16 nats, whereas the largest single-day jump is only 3.2 nats. We therefore also count by *episode*, merging jumps less than 40 days apart: the 158 jumps of the main panel form 65 episodes. Measured this way, the largest shock (15.8 nats, COVID) exceeds what both nulls produce: 8.1 nats on average under the i.i.d. null (none of the 200 repetitions reaches it) and 11.7 under the block null ($p = 0.04$). This result echoes the range of $\mathcal{J}$ and rests on a single episode. The clustering of jumps (2.4 per episode), by contrast, is reproduced by the block null: it is simply the clustering of turbulent days. And the average episode (1.4 nat) is smaller than in both nulls: most real episodes are modest, only a few are exceptional.

Coupling is not just a volatility effect. Three checks, in table 3, rule out the usual objections.

- *Volatility.* Measured correlations rise mechanically when variances rise (Forbes & Rigobon, 2002). We therefore divide each return by its recent volatility before computing correlations ("devolatilisation"). Coupling falls by about 30 % but still passes both nulls on both panels.
- *Extreme days.* We replace each return by its rank, converted to a Gaussian score, which makes the measure insensitive to extreme values. This "rank" version is the most robust of the three.
- *Existing measures.* The absorption ratio of Kritzman et al. (2011), which measures the share of variance explained by the first factors, fails the block null on both panels. The likely reason is that $\log\det R$ accounts for the whole spectrum of the correlation matrix, including the small eigenvalues that collapse when assets couple.

	S&P 500, 20 stocks		FF49, 48 industries	
Measure	i.i.d.	blocks	i.i.d.	blocks
Coupling C, raw	6.2 (< 0.005)	2.0 (0.030)	11.9 (< 0.005)	1.8 (0.025)
Coupling C, devolatilised	5.1 (< 0.005)	2.0 (0.040)	7.9 (< 0.005)	1.7 (0.040)
Coupling C, ranks	9.6 (< 0.005)	**2.3** (**0.030**)	12.7 (< 0.005)	**2.2** (**0.010**)
Absorption ratio	0.7 (0.215)	-0.5 (0.695)	6.9 (< 0.005)	0.4 (0.375)
First-eigenvalue share	5.3 (< 0.005)	1.8 (0.040)	6.7 (< 0.005)	0.2 (0.445)

Table 3: Stress-minus-calm rise of each dependence measure, in standard deviations of the null distribution; in parentheses, the share of the 200 repetitions at least as extreme.

5.2 What does not survive

- *One-sided jumps (H3).* The detector finds 158 upward jumps against 9 downward, and the chart of $\mathcal{J}$ is visibly a sawtooth (figure 2 in the appendix). But the i.i.d. null produces as many: 148 up and 3 down on average, with stronger asymmetry than the data. The sawtooth is what a fat-tailed law produces when passed through the estimator, whether or

not information arrives in bursts. Counting by episode changes nothing: 65 upward episodes against 7 downward, but the nulls are even more one-sided (31 up per down on average under the i.i.d. null, 21 under the block null, against 9 in the data).

- *Shape under stress (H1, $\mathcal{D}$).* Tails thicken under stress (+0.27 nat), but less than in the nulls (+1.25 and +0.65). A "stress" window in a shuffled world contains isolated extreme days, which thicken the measured tails more.
- *Relaxation (H4).* The measured half-life, 121 days, lies between that of the i.i.d. null (92) and that of the block null (163). It is therefore not identified: 121 days cannot be read as an absorption time for shocks. Measured by episode, from the peak until the index is halfway back, relaxation takes 18 days at the median (74 for the five largest episodes), no longer than in the nulls (23 and 22 days; 68 and 107 for the five largest).
- *Sign (H5).* Jumps of $\mathcal{J}$ do coincide with negative skewness (correlation -0.24), but the nulls produce the same effect in 8 to 10 % of cases, above the 5 % threshold. By episode, market skewness falls between the start and the peak in 62 % of episodes, against 50 % under the i.i.d. null ($p = 0.015$) but 53 % under the block null ($p = 0.08$): the sign is still not established.
- *Channel coupling (H6) and compensation (H2).* Dependence and tails move together under stress (0.63 against 0.04 in calm markets), but the i.i.d. null couples them even more (0.90). A window containing a large common day shows both higher correlation and thicker tails, real or shuffled. The same mechanism produces the compensation slope of H2.

5.3 Entropy describes, it does not forecast (H7)

The question is: does knowing today's entropy say anything about risk over the next three weeks that today's volatility does not already say? We compare a model using volatility only with a model that adds an entropy channel. Both are estimated on the past only, re-estimated each year, and compared on dates they have not seen. No channel improves the forecast of volatility, drawdown or worst loss: the best gain is 3.3 % in squared error, with $p = 0.23$. In the estimation sample, by contrast, every coefficient looks highly significant ($p < 10^{-4}$): the classic trap of overlapping windows. Entropy measures are therefore *diagnostic*, not predictive.

5.4 How crises split their intensity (H8)

The previous hypotheses compare averages over a whole regime. H8 looks at each crisis. We merge jumps less than 40 days apart into *episodes*, and split the rise of $\mathcal{J}$ in each episode exactly into dependence, skewness and tails. Skewness almost never dominates (1 episode out of 139). The split between dependence and tails varies: dependence dominates 46 % of the 65 episodes on the main panel, against 26 % on average under the i.i.d. null (range 18–33 %); 79 % of the 61 industry episodes, against 43 %.

Against the block null, however, this result is not identified once blocks reach 21 days (table 12 in the appendix). We therefore draw no conclusion about the nature of crises. We use it for one purpose only, a more modest one: calibrating the composition of a stress scenario (section 6.5), which only requires the observed composition to be more structured than that of a memoryless world. The size of an episode does not predict its composition (rank correlation 0.05).

The takeaway. What distinguishes a real crisis from a series of large days is not the thickening of each asset's tails; it is that assets couple. For a scenario designer, this means that the dependence

structure of a scenario deserves as much attention as the level of the shocks, and that any crisis indicator based on estimated moments should be confronted with a null before use.

6 Application 1: measuring and calibrating scenario severity

This section does not depend on any of H1–H7: it is a measurement convention plus an optimisation. It speaks directly to the design of market scenarios, such as the financial shocks of the European adverse scenario or the global market shocks of US stress tests.

6.1 An entropy-ball scenario

Definition 1 (Scenario of severity η)**.** Let p_0 be the current distribution of returns and ρ a risk measure (for example a portfolio's ES_{99}). The scenario of severity η is the most adverse distribution within η nats of p_0:

$$p^\star_\eta \;=\; \arg \max_{p\,:\,D_{KL}(p\,\|\,p_0)\,\le\,\eta} \rho(p). \tag{3}$$

The construction is known (Breuer & Csiszár, 2013; Glasserman & Xu, 2014). For a portfolio and a Gaussian reference it has an explicit solution: the scenario shifts the portfolio mean down and raises its volatility, both drawn from the same budget of η nats. The simplest case gives the exchange rate between information and risk:

$$\text{a purely directional shock of } \eta \text{ nats is a move of } \sqrt{2\eta} \text{ standard deviations.} \tag{4}$$

One nat buys 1.41 standard deviations, two nats buy two, and a three-standard-deviation shock costs 4.5 nats. The severity parameter is therefore not an abstraction: it translates directly into the language of committees.

6.2 Calibrating the radius on the market itself

It remains to choose η. We read it off the market's information flow, that is, off how much the market revises its own distribution over 21 days, $I_t^{(21)} = D_{KL}(p_t \,\|\, p_{t-21})$, measured on the same object as the constraint (the portfolio's distribution). The historical quantiles of this flow give a radius for each return period (table 4).

Return period	1 yr	2 yr	5 yr	10 yr	20 yr	50 yr†
radius η (nats)	0.017	0.045	0.149	0.408	0.634	1.369
directional shock (std. dev.)	−0.08	−0.13	−0.24	−0.38	−0.47	−0.66
volatility multiplier	1.12	1.20	1.37	1.63	1.80	2.21
one-day ES_{99}	3.9%	4.2%	4.9%	5.9%	6.6%	8.3%
history: realised volatility increase	1.54	1.77	2.28	3.25	3.57	4.43

Table 4: Severity ladder for an equally weighted portfolio of the twenty stocks, calibrated on the 21-day information flow (383 non-overlapping observations). Current state: annual volatility 20 %, one-day ES_{99} 3.4 %. Last row: quantile, at the same return period, of the ratio of the next 21 days' volatility to that of the previous 252 days. † Extrapolated beyond 33 years of data.

The scenario spends its budget mostly on volatility rather than direction: at the ten-year rung, 63 % more volatility and a shift of 0.38 standard deviation, because ES is more sensitive to scale than to location. The radius depends on the portfolio: a ladder must be calibrated for the portfolio it will be applied to.

A serious limitation. The last row of the table confronts the ladder with history. A ten-year scenario multiplies volatility by 1.6, whereas the ten-year quantile of the realised volatility increase is 3.2. The ladder is therefore *too mild* by a factor close to two. Either the information flow, measured on smoothed estimators, understates the true revision; or a Gaussian ball is not the right geometry for volatility shocks this abrupt. We do not decide between the two. The return periods in the table should therefore not be quoted as frequencies. What survives this limitation are the *relative* comparisons of the next two sections.

6.3 Auditing an existing scenario

The same unit prices a scenario built by expert judgement. Take the type of scenario common in stress exercises: every volatility multiplied by the same factor, every correlation forced to a high value, and a directional shock of several standard deviations. Its cost in nats, $D_{KL}(p_{\text{scenario}} \,\|\, p_0)$, is available in closed form. It can then be compared with the entropy-ball scenario that produces *the same tail loss* (table 5).

Classical scenario	moderate	standard	severe
volatility multiplier	1.5	2.0	3.0
all correlations forced to	0.70	0.90	0.95
directional shock	-2σ	-3σ	-4σ
resulting one-day ES_{99}	9.1%	13.7%	20.3%
cost seen from the portfolio (nats)	2.75	7.24	16.22
cost of the entropic scenario with the same ES_{99}	**1.85**	**5.47**	**13.74**
overpayment of the classical scenario	33 %	24 %	15 %
Cost of each component alone, on the 20 assets (nats)			
directional shock	2.00	4.50	8.00
volatility dilation	4.39	16.14	58.03
forced correlations	2.43	8.10	13.50
full scenario	6.80	9.97	20.19

Table 5: Cost in nats of classical scenarios, compared with the entropic scenario that produces the same tail loss. The "cost seen from the portfolio" is the divergence on the portfolio's distribution, the only one comparable with the radius of a ball. For reference, the largest radius observed in 33 years is about 2 nats, and the worst day in the sample costs 9 nats.

Three readings.

- *The classical scenario is too expensive.* For the same tail loss it costs 15 to 33 % more nats than necessary (6 to 15 % on French's industries). The reason is structural: it also distorts directions the portfolio does not see.
- *The expensive component is the uniform dilation of volatilities.* Shifting the mean by three standard deviations costs 4.5 nats. Doubling every volatility costs 16.1 nats on its own, more than the whole scenario.
- *Volatility and correlation pay for each other.* The three components cost 28.7 nats separately but 10.0 together. Forcing correlations up concentrates risk on a single market factor, and dilating that factor is far cheaper than dilating every direction. This is exactly what observed crises do: they load risk onto a few common factors (section 5). A scenario that shocks volatility and correlation as two separate dials cannot represent it.

For an authority, the practical value is the ability to *audit* an existing scenario without replacing it: each scenario receives a cost in nats, comparable across exercises and asset classes, and one can check that it does not buy its severity through a needlessly improbable route.

6.4 Example: the EBA 2025 adverse scenario in nats

We apply the measure to the adverse scenario of the 2025 EU-wide stress test, designed by the ESRB and published by the EBA (EBA, 2025). Its first year, the most severe, cuts EU stock prices by 50%, raises long-term rates by 110 basis points in Germany and 168 for the EU, and leaves the euro's exchange rates unchanged. Read as a shift of the mean of one-year changes, this shock costs $\frac{1}{2}\Delta^\top\Sigma^{-1}\Delta$ nats, where Σ is the historical covariance of the same variables in the euro-area panel (full calculation and table 9 in appendix C).

The scenario costs 10.5 nats, the equivalent of a 4.6-standard-deviation shock on a single factor. It is therefore *severe but plausible* in a precise sense: as improbable as 2008 (11.2 nats) or 2022 (11.9) as they actually unfolded, less than the worst observed year (14.4, March 2008 to March 2009), far more than 2011 (3.1). It uses dependence well: taken one by one, the five indices and two rates would cost 31.9 nats, but national indices move almost as a single factor (average correlation 0.87). Finally, the choice of Σ is a choice of severity: with a crisis-period covariance (2008–2012), the same scenario costs only 8.4 nats. The measure makes that choice explicit.

6.5 Composing a scenario: volatility or correlation?

The entropy ball fixes the severity *of the portfolio* (its stressed mean and volatility). It does not say how each asset's covariance should move to get there. The default used so far is a uniform dilation of volatilities with correlations unchanged. That is a convention, not a fact about crises. Yet section 5 shows that, under stress, dependence moves on its own.

A simple property lets us account for this without changing severity. A portfolio's variance, $w^\top\Sigma w$, is linear in Σ. If two covariances give the same portfolio variance, any mixture of the two does too (proposition in appendix E). One can therefore mix an "all-volatility" covariance and an "all-correlation" covariance with a weight α, the *dependence share*, without touching severity. One caveat: correlation alone cannot carry any severity. With every correlation equal to one, our portfolio's volatility cannot exceed 1.5 times its current level; beyond that, the rest comes from dilation.

The dependence share is calibrated on the robust results. The *anchor* comes from H1: under stress, dependence accounts for $0.99/(0.99 + 5.97) = 14\,\%$ of the move in Gaussian entropy (scale plus dependence). The *refinement* comes from H8: on the more severe half of episodes, that share reaches 49 %.

Composition of the ten-year scenario ($\eta = 0.41$)	dependence share	total cost (nats)
default: uniform dilation	0.00	6.80
calibrated on H1 (regime average)	0.14	2.88
refined by H8 (severe episodes)	0.49	0.71

Table 6: Total cost in nats, on the twenty assets, of three compositions that produce exactly the same portfolio severity ($\mathrm{ES}_{99} = 5.9\,\%$). On French's industries: 14.7, 8.3 and 5.8 nats for shares of 0, 0.20 and 0.75.

The result runs against intuition (table 6): the larger the dependence share, the *less* improbable the scenario, at equal severity. Uniform dilation is therefore not a prudent choice; it is both less

realistic and costlier in plausibility than a composition anchored on history. This finding is specific to this panel and should not be read as a general law.

6.6 Reverse stress testing

The same unit also answers the reverse question: what is the least improbable scenario that produces a given loss? On the main panel, the daily loss exceeded one day in a hundred (3.2 %) costs 1.5 nats; the one exceeded one day in a thousand (7.7 %) costs 4.8 nats; the worst day in 33 years (11.5 %) costs 9.0 nats. Beyond a few nats, the non-parametric version of this calculation rests on a handful of days only, and the Gaussian version should be preferred.

7 Application 2: an entropic CISS

The paper's only robust result, coupling under stress, becomes here an indicator in the format of the ECB's CISS (Holló, Kremer & Lo Duca, 2012). Like the CISS, it covers the money-market, bond, equity, financial-intermediary and foreign exchange segments. Unlike the CISS, it says what part of the coupling lies within each segment and what part lies *between* segments. For a scenario designer, this is precisely the information needed to set the correlations between shocks to different markets.

7.1 An exact decomposition by segment

Split the series into segments $B_1, \ldots, B_K$. Total coupling decomposes exactly:

$$C(R) \;=\; \underbrace{\sum_{k=1}^{K} C(R_{kk})}_{\text{within segments}} \;+\; \underbrace{I(B_1;\ldots;B_K)}_{\text{between segments}}, \tag{5}$$

where R_{kk} is the correlation matrix of segment k. The between-segment term is zero exactly when segments move independently of one another: it is the systemic part. For two segments A and B, the mutual information $I(A;B)$ gives a contagion map in the same unit. Everything is unit-free: equity returns, rate changes and exchange rates can sit in the same matrix. We use the rank version of coupling.

7.2 The euro-area panel

The panel is daily, from January 2005 to October 2026 (5,690 days), with 28 series: the overnight rate (EONIA, then €STR) for the money market; six ECB sovereign yields and spreads for bonds; five national indices for equities; nine banks and three insurers for intermediaries; four euro exchange rates. For the money market the CISS uses the Euribor minus overnight spread; since Euribor is not available on the ECB data portal, this segment currently contains the overnight rate only.

7.3 Results

Three readings (table 7 and figure 1).

- *Stress ties segments together*, beyond both nulls. Every pair outside the money market rises; the links between equities and intermediaries and with FX are the clearest.

Component	Rise under stress (nats)	p, i.i.d.	p, blocks
Total coupling	+1.61	< 0.005	< 0.005
Between segments	**+0.19**	< 0.005	$<$ **0.005**
Within: equities / intermediaries	+0.46 / +0.48	< 0.005	< 0.005 / 0.045
Within: bonds / FX	+0.34 / +0.14	< 0.005	< 0.005
Equities – intermediaries	+0.096	< 0.005	0.025
Bonds – FX	+0.055	< 0.005	< 0.005
Bonds – intermediaries	+0.040	< 0.005	0.050
Money market – other segments	≈ 0	n.s.	n.s.

Table 7: Euro area: stress-minus-calm rise against the two nulls (200 repetitions each). Stress is the top decile of the equity block's volatility, threshold computed on the past only.



Figure 1: The euro-area entropic CISS. Top: total coupling and the ECB's CISS, standardised. Middle: coupling between the five segments, in nats, and its share of the total. Bottom: mutual information between pairs of segments. Grey areas: CEPR recessions.

- *The sovereign–bank link stands out.* Mutual information between bonds and intermediaries peaks at the end of 2011, at the height of the sovereign debt crisis (0.46 nat against 0.10 at the end of 2006). The aggregate CISS cannot show which segments an episode ties together.
- *The money market does not couple* in daily data: the overnight rate follows monetary policy, not market stress.

	Total coupling	Between segments	Volatility	CISS
Correlation with the CISS	0.58	0.35	0.74	1
AUROC, CEPR recessions	**0.85**	0.75	0.80	0.83
Explains the CISS beyond volatility (t)	**6.7**	1.7		
Growth-at-Risk, alone: p	0.04	< 0.001	0.98	< 0.001
Growth-at-Risk, with CISS and past growth: p	0.78	0.81		< 0.001

Table 8: Euro area against official indicators. AUROC is the probability that a randomly drawn recession date has a higher indicator than a non-recession date (0.5: no power; 1: perfect separation). Growth-at-Risk: regression of the lower decile of four-quarter GDP growth on the indicator (Adrian, Boyarchenko & Giannone, 2019), 79 quarters.

Against official indicators (table 8), total coupling flags recessions slightly better than the CISS and than volatility, and it explains the CISS beyond what volatility explains. It does not, however, forecast growth once the CISS is accounted for. With three recessions and 79 quarters, these comparisons remain indicative. The indicator is therefore *coincident*: a complement to the CISS that says which segments are linked, not a better predictor. On US data, against the St. Louis Fed financial stress index (Kliesen & Smith, 2010), coupling adds to volatility on the twenty stocks ($t \approx 4$ to 5), mostly before 2015, but not on the 48 industries.

Next steps. Add Euribor from its administrator (EMMI) to rebuild the money-market segment as in the CISS; date stress episodes with the ESRB crises database (Lo Duca et al., 2017) and measure how the contagion map behaves around them; confront the indicator with institution-level systemic risk measures, SRISK (Brownlees & Engle, 2017) and ΔCoVaR (Adrian & Brunnermeier, 2016).

8 Recommendations and limitations

8.1 Recommendations for scenario designers

1. *Price every scenario in nats.* The cost $D_{KL}(p_{\text{scenario}} \,\|\, p_0)$ is available in closed form for a Gaussian scenario. It makes scenarios on different markets and in different exercises comparable, and it reads in standard deviations (k standard deviations $= k^2/2$ nats).
2. *Check efficiency before severity.* Compare a scenario's cost with that of the entropic scenario with the same loss: an overpayment of 15 to 33 %, as for the classical scenarios tested here, signals severity bought through a needlessly improbable route.
3. *Do not dilate volatilities uniformly by default.* Shocking part of the severity through dependence is both more faithful to observed crises and less improbable at equal severity.
4. *Set cross-market correlations from a coupling map.* The decomposition (5) says which segments stress ties together and by how much, in a common unit.

5. *Run a placebo null* before adopting any crisis indicator based on estimated moments.

8.2 Limitations

- *Few results survive.* Only coupling under stress passes both nulls on every panel. Jumps, relaxation and sign are not established.
- *Return periods are not validated.* The severity ladder is about twice too mild compared with history (table 4). Only relative comparisons (overpayment, composition) withstand this flaw.
- *H8 is retained only against the i.i.d. null.* It serves only to calibrate a composition and should not be used to tell crisis types apart.
- *Nothing here forecasts.* The measures are diagnostic.
- *Shape is estimated asset by asset.* Non-linear tail dependence is left out of $\mathcal{J}$, which therefore understates the true divergence, especially under stress.
- *Cross-sections are small.* Twenty stocks or 48 industries are not enough to make the figures presented constants; a scenario radius is not comparable from one asset universe to another.
- *The entropic CISS is incomplete.* Its money-market segment awaits Euribor, and its comparisons rest on three recessions.

9 Conclusion

If the Gaussian is the distribution of maximal ignorance, any departure from it measures information, in nats. This idea gives an exact decomposition of a market's uncertainty into three channels, and a unit-free index, $\mathcal{J}$, that measures the distance from maximal ignorance.

The empirical result is first a warning. Every prediction of the framework is visible in the raw data: the index rises under stress, it moves as a sawtooth, its jumps are one-sided and come with negative skewness. Rerun on data whose time order has been shuffled, almost all of it disappears, and several effects are even stronger in the shuffled data. Only one thing durably separates a crisis from a run of large days: markets couple.

For institutions that design stress scenarios, the framework offers three tools. A unit, the nat, that measures a scenario's distance from the current state of the market and reads in standard deviations. An audit, which shows that classical scenarios pay 15 to 33 % too much for their tail loss, because they dilate volatilities uniformly instead of coupling markets as crises do. And a map, the entropic CISS, that says which segments stress ties together: a complement to the CISS, not a predictor.

A The bridge from order flow to Gaussian prices

Over an interval, let V be traded volume and Q net signed order flow (buys minus sells). Two assumptions suffice. *Linear impact*: the log-price change equals λQ, as in Kyle (1985). *Variance proportional to volume*: given V, the market knows only the mean of Q and a variance $\varsigma^2 V$. Maximum entropy then gives

$$\Delta X \mid V \;\sim\; \mathcal{N}\big(\lambda m,\; \lambda^2 \varsigma^2 V\big)\,. \tag{6}$$

The conditioning variable is volume, not time: this is the mixture-of-distributions hypothesis of Clark (1973) and Tauchen & Pitts (1983), and Ané & Geman (2000) show that returns measured in

transaction time are close to Gaussian. Unconditionally, the return is a mixture of Gaussians, hence fat-tailed (Mandelbrot & Taylor, 1967): fat tails alone do not prove that information is present. If, in addition, increments are independent in the volume clock, the log-price is a time-changed Brownian motion and the price a geometric Brownian motion. Both assumptions are imperfect (actual impact is closer to a square root, and order flow is autocorrelated), which is why all empirical work is done in calendar time.

B The dynamics of the index: a theorem and two postulates

The theorem. For a return X and a signal Y, $\mathbb{E}_Y[h(X \mid Y)] = h(X) - I(X;Y) \leq h(X)$ (Cover & Thomas, 2006). Receiving information lowers entropy *on average* by exactly the mutual information. A particular realisation can raise it (a confusing signal); the jump component is therefore mainly, not exclusively, one-sided.

Postulate 1 (relaxation). Absent new information, $\mathcal{J}$ returns to a long-run level: $d\mathcal{J}_t = -\kappa(\mathcal{J}_t - \bar{\mathcal{J}})\,dt + \sigma_{\mathcal{J}}\,dW_t$. An Ornstein–Uhlenbeck process is the only process that is Markov, Gaussian and stationary at once (Doob, 1942). Adding the jumps caused by news gives

$$d\mathcal{J}_t \;=\; -\kappa(\mathcal{J}_t - \bar{\mathcal{J}})\,dt \;+\; \sigma_{\mathcal{J}}\,dW_t \;+\; dZ_t, \tag{7}$$

where Z_t is the sum of positive jumps up to t: the class of Barndorff-Nielsen & Shephard (2001). The half-life of a shock is $\ln 2/\kappa$.

Postulate 2 (sign). $\mathcal{J}$ is an even function of skewness: a distribution with a long left tail (a crash) and its mirror image (a rally) have the same $\mathcal{J}$. To say anything about direction, an outside ingredient is needed. We postulate that jumps come with negative skewness, relying on the leverage effect (Black, 1976; Engle & Ng, 1993).

Why dependence and tails should rise together. Important news hits every asset on the same day. On that day assets move together (coupling rises) and each records an unusual loss (its tails thicken). The two channels describe the same day seen from two places. In calm markets, large moves are asset-specific and couple nothing. This is H6. The null shows, however, that this coincidence is also produced mechanically by estimation.

C Computing the cost in nats of the EBA 2025 adverse scenario

Code: scripts/eba_nats.py; results: paper/tables/eba_nats.csv.

1. The shocks. We take the adverse macro-financial scenario (EBA, 2025) for 2025, as a deviation from the baseline (annual averages): EU stock prices −50%, the baseline assuming them unchanged (table 4.1.7 of the document); long-term rates for Germany 3.50% against 2.40% and for the EU 4.80% against 3.12% (table 4.1.6); euro exchange rates against the dollar, pound, yen and Swiss franc unchanged (table 4.1.10). Each shock is mapped to the matching series of the euro-area panel (section 7), giving an eleven-component vector Δ:

Cost in nats	equities and rates	with FX held fixed
EBA 2025 adverse scenario, first year	**10.5**	**11.1**
equities only / rates only	6.8 / 3.1	
sum of variable-by-variable costs	31.9	
Observed years: 2011 / 2008 / 2022	3.1 / 11.2 / 11.9	3.1 / 18.2 / 13.2
worst rolling year (March 2008–March 2009)	14.4	19.5

Table 9: Cost in nats of the EBA 2025 adverse scenario and of observed years, on five equity indices, two ten-year yields and four exchange rates, annual covariance estimated on 2004–2024.

EBA shock	Panel series	Δ_i	σ_i (1 year)	cost alone (nats)
EU stocks −50%	DAX, CAC 40, FTSE MIB, IBEX 35, AEX	$\ln 0.5 = -0.693$	19–23%	4.4–6.5
German long rate	ECB AAA 10-year yield	+1.10 pt	0.65 pt	1.4
EU long rate	ECB all-ratings 10-year yield	+1.68 pt	0.68 pt	3.0
FX unchanged	EUR/USD, GBP, JPY, CHF	0	7–11%	0

Equities enter as log returns, rates as level changes; the document does not state the maturity of long-term rates, which we take as ten years. GDP, unemployment, real estate and inflation, which are not market prices, are not used.

2. The reference. $p_0 = \mathcal{N}(0, \Sigma)$, where Σ is the covariance of one-year changes of the eleven series, estimated on daily changes from 6 September 2004 to 31 December 2024 (the data available to the designer) and multiplied by 252.

3. The formula. The scenario fixes means, not volatilities or correlations. We read it as a pure mean shift, hence (appendix E) $D_{KL}(p_{\text{scenario}} \,\|\, p_0) = \frac{1}{2}\Delta^\top \Sigma^{-1} \Delta$. This is the *minimum* cost compatible with the scenario: among all distributions with mean Δ, the shifted Gaussian is closest to p_0. Variables the scenario does not fix (two- and five-year yields, banks, etc.) are left free; completing them with their conditional expectation is cheapest, and the cost then depends only on the shocked block S: $\frac{1}{2}\Delta_S^\top \Sigma_{SS}^{-1} \Delta_S$. The cost of a single variable is $\frac{1}{2}(\Delta_i/\sigma_i)^2$; for the DAX ($\sigma = 20.3\%$), the shock is 3.4 standard deviations and costs 5.8 nats. η nats read as $\sqrt{2\eta}$ standard deviations.

4. Reading. The variable-by-variable costs sum to 31.9 nats, the joint cost is 10.5: a common shock to highly correlated indices follows the most probable direction of the data. Equities alone (6.8) plus rates alone (3.1) leave 0.5 nat of interaction (falling equities with rising rates go against the historical correlation, +0.13). Holding FX fixed adds 0.7 nat.

5. Observed years. Same calculation, same Σ, on the realised changes of the eleven series (31 December to 31 December; worst 252-day window ending 17 March 2009). The median year costs 4.1 nats. The scenario combines the equity fall of 2008 (when rates fell) and the rate rise of 2022 (nearly 3 points).

6. Variants (equities and rates / with FX fixed). Covariance of overlapping annual changes: 12.8 / 18.7; sample to 2026: 10.6 / 11.3; crisis covariance 2008–2012: 8.4 / 9.0. At the end of the three-year horizon (2027: −42%, +87 and +112 bp, three-year covariance), the cost falls to 2.1 / 2.2: the severity is concentrated in the first year.

7. Limits. Shocks are annual averages; the EU equity fall is applied to each national index; the Gaussian reference has no fat tails, which tends to overstate the cost. The ECB's instantaneous market-risk scenario is not priced (detailed tables not accessible), but the same formula applies. These costs (eleven series, one year) are not directly comparable to table 4 (equity portfolio, 21 days).

D Empirical details

Estimators. Two covariance estimators run in parallel: a 252-day rolling window with the Ledoit & Wolf (2004) correction, and exponential weighting with a 60-day half-life and 10 % shrinkage towards the diagonal. The second prevents an extreme day *leaving* the window from creating a spurious jump. The shape deficit uses the Hyvärinen (1998) approximation, whose two terms separate a skewness channel and a tail channel; the better-known Edgeworth expansion $S^2/12 + K^2/48$ overstates the deficit of a Student law with 4 degrees of freedom by a factor of 69. Jumps are flagged at four times a local bipower-variation scale lagged by one day (Barndorff-Nielsen & Shephard, 2004).

Inference. Two consecutive rolling windows share 251 of their 252 days. Regime comparisons therefore use a 252-day circular block bootstrap. The relaxation p-value follows the Dickey–Fuller distribution. Forecasts (H7) are re-estimated each year on the past, compared on dates 21 days apart, with the Diebold & Mariano (1995) test.

Raw stress-regime results (S&P 500). Under stress (1,076 dates against 6,481), scale rises by 5.97 nats, dependence falls by 0.99, shape rises by 0.27 ($p = 0.16$), $\mathcal{J}$ rises by 1.94 ($p < 0.001$, but without passing the nulls). Average correlation goes from 0.26 to 0.32, the effective number of modes from 13.1 to 11.4. Market skewness does not change significantly between regimes: the skewness signature is a property of jumps, not of the regime. The index shares about a fifth of its variance with volatility ($r = 0.46$), and the shape channel is orthogonal to it ($r = 0.08$).

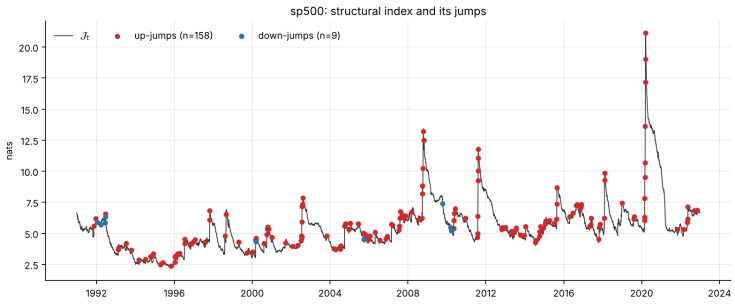


Figure 2: The structural index $\mathcal{J}_t$ on twenty S&P 500 stocks, 1990–2022, with detected jumps (red: up, blue: down). The sawtooth is visible, but the i.i.d. null reproduces it.

E Essential proofs

Gaussian entropy and equation (1). For $q = \mathcal{N}(\mu, \Sigma)$, $h(q) = \frac{1}{2}\log\big((2\pi e)^N \det\Sigma\big)$. With $\Sigma = DRD$, $\det\Sigma = (\prod_i \sigma_i)^2 \det R$, hence $h(q) = \frac{N}{2}\log(2\pi e) + \sum_i \log\sigma_i + \frac{1}{2}\log\det R$. Since $\det R$ is the product of the eigenvalues of R, which sum to N, the arithmetic–geometric mean inequality gives $\det R \leq 1$: correlation always destroys entropy.

			i.i.d. null		21-day block null		
	Statistic	Obs.	mean	5–95 %	mean	5–95 %	Verdict
H1	gap H^{dep}	−0.99	−0.21	−0.42–−0.00	−0.51	−0.90–−0.17	**passes both**
H1	gap $\mathcal{J}$	1.93	2.14	1.59–2.79	1.97	1.18–2.90	fails both
H1	gap $\mathcal{D}$	0.27	1.25	0.82–1.75	0.65	0.20–1.19	wrong side (i.i.d.)
H2	corr($\Delta H^{vol}, \Delta H^{dep}$)	0.10	0.23	−0.26–0.60	0.27	−0.21–0.63	fails both
H2	β (stress)	0.20	0.05	−0.33–0.31	0.11	−0.28–0.32	fails both
H3	positive jumps	158.0	148.5	136.9–161.0	133.2	121.0–146.0	block only
H3	negative jumps	9.0	3.0	1.0–6.0	3.7	1.0–7.0	wrong side (both)
H3	mean positive jump (nats)	0.62	1.12	0.95–1.29	0.84	0.69–1.03	wrong side (both)
H3	skewness of $\Delta\mathcal{J}$	12.26	14.80	12.27–17.60	12.97	9.71–17.65	wrong side (i.i.d.)
H3	var. share of pos. jumps	0.838	0.899	0.869–0.922	0.817	0.764–0.860	wrong side (i.i.d.)
H4	half-life (days)	120.8	92.1	82.4–103.3	162.8	116.0–227.8	between the two nulls
H4	sd($\mathcal{J}$)	2.03	1.55	1.31–1.81	2.06	1.40–2.75	i.i.d. only
H4	range($\mathcal{J}$)	18.95	10.05	8.05–12.88	13.77	9.04–17.88	**passes both**
H5	gap ΔS_{mkt}	−0.053	−0.011	−0.088–0.070	−0.027	−0.082–0.039	fails both
H5	corr($\Delta\mathcal{J}, \Delta S_{mkt}$)	−0.243	−0.035	−0.298–0.230	−0.069	−0.268–0.209	fails both
H6	coupling, stress	0.63	0.90	0.84–0.94	0.72	0.56–0.82	wrong side (i.i.d.)
H6	coupling gap	0.590	0.769	0.671–0.871	0.668	0.513–0.822	wrong side (i.i.d.)
–	corr($\mathcal{J}$, log vol)	0.461	0.414	0.359–0.466	0.264	0.164–0.362	—

Table 10: S&P 500: observed statistics against the two nulls, 200 repetitions each. "Wrong side": the null exceeds the data. Gaps: stress minus calm.

			i.i.d. null		21-day block null		
	Statistic	Obs.	mean	5–95 %	mean	5–95 %	Verdict
H1	gap H^{dep}	−3.65	−0.56	−1.03–−0.19	−2.46	−3.45–−1.41	**passes both**
H1	gap $\mathcal{J}$	5.93	7.95	6.12–10.11	7.79	5.37–10.39	wrong side (i.i.d.)
H1	gap $\mathcal{D}$	0.98	4.81	3.47–6.39	2.60	1.15–3.99	wrong side (both)
H2	corr($\Delta H^{vol}, \Delta H^{dep}$)	−0.63	−0.36	−0.54–−0.19	−0.54	−0.65–−0.41	i.i.d. only
H2	β (stress)	0.23	0.08	−0.05–0.21	0.28	0.15–0.41	i.i.d. only
H3	positive jumps	182.0	181.8	165.0–201.1	176.5	159.0–194.1	fails both
H3	negative jumps	3.0	7.8	4.0–13.0	5.2	2.0–10.0	i.i.d. only
H3	mean positive jump (nats)	1.70	3.47	2.74–4.18	2.30	1.82–2.90	wrong side (both)
H3	skewness of $\Delta\mathcal{J}$	17.04	18.31	14.84–21.83	16.46	11.76–26.04	fails both
H3	var. share of pos. jumps	0.889	0.943	0.918–0.958	0.877	0.840–0.910	wrong side (i.i.d.)
H4	half-life (days)	189.8	101.0	89.7–112.1	199.3	136.6–273.6	between the two nulls
H4	sd($\mathcal{J}$)	6.37	5.46	4.43–6.48	7.05	5.05–9.18	fails both
H4	range($\mathcal{J}$)	54.47	36.86	28.53–43.95	45.86	32.98–59.10	i.i.d. only
H5	gap ΔS_{mkt}	−0.112	−0.057	−0.140–0.017	−0.068	−0.123–−0.010	fails both
H5	corr($\Delta\mathcal{J}, \Delta S_{mkt}$)	−0.440	−0.197	−0.472–0.075	−0.177	−0.441–0.149	fails both
H6	coupling, stress	0.76	0.94	0.91–0.96	0.82	0.74–0.87	wrong side (i.i.d.)
H6	coupling gap	0.411	0.635	0.510–0.757	0.558	0.390–0.717	wrong side (i.i.d.)
–	corr($\mathcal{J}$, log vol)	0.677	0.443	0.388–0.490	0.327	0.240–0.409	—

Table 11: FF49 industries, 1990–2026: same conventions as table 10.

Block length (days)	null mean	5–95 %	Verdict
1 (i.i.d.)	0.26	0.18–0.33	passes
5	0.31	0.23–0.41	passes
10	0.35	0.27–0.45	marginal ($p = 0.08$)
21	0.43	0.33–0.54	fails
42 / 63 / 126	0.49 / 0.51 / 0.47	0.40–0.60	fails

Table 12: H8 on the S&P 500: observed share of dependence-dominated episodes (0.46) against the block null, by block length.

The deficit is a divergence. If q is the Gaussian with the mean and covariance of p, then $\mathbb{E}_p[\log q] = \mathbb{E}_q[\log q]$ because $\log q$ is quadratic. Hence $D_{KL}(p \,\|\, q) = h(q) - h(p) = \mathcal{D} \geq 0$, with equality if and only if p is Gaussian.

The structural index is a divergence. With q_i the Gaussians with each asset's mean and volatility, $D_{KL}(p \,\|\, \prod_i q_i) = -h(p) + \sum_i \log \sigma_i + \frac{N}{2} \log(2\pi e)$. Substituting (1) for $h(p)$ leaves $\mathcal{D} - \frac{1}{2} \log\det R = \mathcal{D} + C = \mathcal{J}$.

Mixing compositions. If $w^\top \Sigma_a w = w^\top \Sigma_b w = s^2$, then for every $\alpha \in [0, 1]$, $w^\top\big((1-\alpha)\Sigma_a + \alpha\Sigma_b\big)w = s^2$, by linearity.

The price of a directional shock. For two Gaussians with the same covariance, $D_{KL}(p_1 \,\|\, p_0) = \frac{1}{2}\Delta^\top \Sigma^{-1} \Delta$ where Δ is the difference in means. A shock of k standard deviations on a portfolio therefore costs $k^2/2$ nats, and η nats buy $\sqrt{2\eta}$ standard deviations.

Decomposition by segment. Applying the definition of C to each block,

$$C(R) - \textstyle\sum_k C(R_{kk}) = \frac{1}{2}\big(\sum_k \log\det R_{kk} - \log\det R\big),$$

which is the Gaussian mutual information between blocks, non-negative by Fischer's inequality.

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